The Influence of Gender Stereotypes on Quantifiers

Alessio Novel

University of Trento

alessio.novel@studenti.unitn.it

1. Aim of the Research

The topic of this proposal focuses on the last of the three levels of analysis we've seen throughout the course: the psychological one. Studies at this level aim to understand how "meaning" is represented and processed in different individuals and why.

A well-known challenge in linguistic theory is to explain how speakers and listeners successfully communicate with quantifiers such as *few* and *many*, even though their meaning is so vague and context-dependent [1]. For example, the numerical value a person gives to the sentence "many children" is different from the numerical value the same person gives to "many points in an NBA game" (the first will likely be 4 or 5, while the second will be around 120).

The fact that the meaning of *many* is context-dependent is not a problem for communication, assuming that people agree on the different meanings the word assumes in different contexts. The problem is that differences in meaning are present even within the same context, which leads to practical problems in communication. This means that there are also some non-contextual factors that shape the meaning of quantifiers.

I hypothesize that one of these factors could be the already well-known issue of stereotypes, which are prevalent in our society. Understanding and quantifying the consequences of stereotypes could lead to a better understanding of how factors like these should be taken into account when studying the semantics of quantifiers.

2. Prior Research

It is already established that the semantics of quantifiers derive from comparisons with prior expectations based on the context [2] [3]. For example, participants change their judgments about the appropriateness of a quantifier for the same numerosity of objects (e.g., balls) depending on the context in which these objects occur (e.g., balls presented in a small versus a large bowl) [4].

A common practice in the study of the semantics of quantifiers is to determine the thresholds that make a quantifier "true" for a person or a group of people. In particular, it has been found that the quantifier *many* has various thresholds depending on the context in which it is used [5].

A relevant type of analysis is to check if participants can form clusters that explain the different possible interpretations a quantifier can assume. This analysis allows us to understand if multiple interpretations are possible and to analyze the factors that induce a person to use one specific interpretation. For example, people were successfully divided based on the way they understand the quantifier *some*, which can be interpreted as "some and possibly all" (logical interpretation) or "some but not all" (pragmatic interpretation) [6].

Individual differences in the semantics of quantifiers were successfully identified using a computational model that separates (using different parameters) the threshold, the vagueness, and the error response for each quantifier for each participant [7] [8]. These studies used real English quantifiers alongside six-letter pseudowords to ensure that participants' judgments were based solely on the meaning of the quantifiers and numerical proportions, thereby neutralizing the effect of context. However, the explanation of why these individual differences arise was not tackled.

In summary, numerous studies highlight individual differences in the interpretation of quantifiers, but the factors that drive these differences have not been studied specifically yet. While there are studies focusing on individual differences in other types of words, such as adjectives and nouns [9] [10], this is not yet the case for quantifiers.

One study that analyzed stereotypes in quantifier exists, regarding temporal quantifiers (such as *frequently*). In the experiment, they used sentences like "The grandfather frequently writes a letter to communicate with family and friends" and "The teenager frequently writes a letter to communicate with family and friends". The problem is that they did not make a differentiation between stereotyped individuals and non-stereotyped individuals, and the result was a general and systematic tendency across all participants. In my opinion, this could be simply attributed to the context (the context of a grandfather versus the context of a teenager) instead of a stereotype contribution.

3. Key Questions

As previously stated, research in this field has primarily focused on how context changes the semantics of quantifiers. In this proposal, I will focus on the variations that appear within the same context. The only previous research that studied individual differences within the same context [7] [8] used English pseudowords to ensure there was no influence from the context. In my opinion, this might not be the case: it is possible that the individual differences found were due to different people associating or simulating the pseudowords within real-world contexts in different ways. This would mean that individual differences are not merely matters of quantifier interpretation, but rather result from the different ways individuals concretize an abstract context. Therefore, I believe it is better to utilize a fixed, real-world context to ensure that the context is truly the same in every participant's mind.

The main aim of my proposal is to identify the non-contextual factors that create differences in interpretation between individuals. My hypothesis is that the well-known presence of stereotypes is a factor that guides this difference.

The core issue is whether, and to what extent, stereotyped individuals interpret quantifiers differently in a stereotyped context.

The first question is whether the stereotype a person holds relates to the value assigned to a quantifier in a given context. We expect that a stereotyped person in a pro-stereotyped context will assign a higher (or lower, depending on the type of

stereotype) value compared to other people in the same context. In other words, I want to see if stereotypes can influence the semantics of quantifiers in a deterministic way.

The second question is whether, following the clustering already performed in other studies [6], it is possible to cluster people into "stereotyped" and "non-stereotyped" groups that lead to different interpretations of the same quantifier. In this case, there may not be only two interpretations, but potentially a spectrum of interpretations depending on the degree of stereotype a person holds; nevertheless, clustering would still be possible.

We will focus on the quantifier *many*, which is frequently taken into consideration because it is known to elicit individual differences due to its vagueness. Its threshold is typically around 40–50% (the center of the scale), which is optimal for studying variation compared to quantifiers at the extremes of the scale. While the study could be augmented by using sentences with the quantifier *few*, it is well known that *few* suffers from polarity effects (such as higher processing costs and response errors); therefore, I have decided to limit the analysis to *many*.

Consequently, the practical research questions are:

- Do stereotyped people have a different threshold for the word *many* compared to non-stereotyped people in a stereotyped context?
- Do people cluster into two groups based on how they interpret the quantifier in a stereotyped context, and does this division reflect the distinction between stereotyped and non-stereotyped individuals?

4. Intended Results

If we can successfully demonstrate that the differing interpretations of quantifiers are dependent on the stereotypes a person holds, and that it is possible to cluster individuals based on these stereotypes to predict their interpretation, we will demonstrate that factors like bias must be taken into account in studies of quantifier semantics. This would represent a significant step toward explaining why individual differences arise in the interpretation of these terms.

We expect that stereotyped individuals in stereotyped contexts will exhibit lower or higher thresholds (depending on the type of stereotype) compared to non-stereotyped individuals in the same contexts.

Obviously, if these differences arise, they represent a significant challenge for communication that must be taken into consideration. Furthermore, by linking this with studies that address the origin of these stereotypes, we can work toward making the semantics of quantifiers more consistent across individuals, ultimately striving for a world where people communicate more effectively. This result would increase public awareness regarding the fact that individuals interpret "objective" words like quantifiers differently depending on their biases.

In the event that a stereotypical influence does not emerge, further studies will be required to definitively test the hypothesis that stereotypes do not influence the semantics of quantifiers (perhaps using different approaches to ensure the design of this study was not insufficient). Conversely, if subsequent studies confirm that stereotypes do not influence quantifiers, we can exclude this factor from future research and focus on other variables to explain individual differences in interpretation.

In either case, this study represents an important starting point for understanding the factors that drive individual differences in the interpretation of quantifiers.

While this study will specifically utilize gender stereotypes, exploring the differences between various types of stereotypes remains an interesting path for future research.

5. Procedure

The quantifier *many* can have two different interpretations [5]: the "cardinal" interpretation, which focuses on the exact number of entities, and the "proportional" interpretation, which focuses on the percentage of entities relative to the total number. I have decided to force the proportional interpretation (which is the more common one) to avoid ambiguities that could arise if a participant uses a cardinal interpretation, as this could alter the results. Using the construction "Many of the..." forces the proportional reading.

I also decided to use gender stereotypes related to occupations in the experiment because they are prevalent, and there are already existing tests and studies regarding them [11].

My proposal is to first determine the level of gender stereotype each participant holds using established tests (such as the IAT Gender-Career test or a general Likert-scale questionnaire) to establish a sort of expected prior for each participant.

Like many psychological concepts, semantic representations are latent. This means that we cannot measure them directly. Instead, we must infer them from observable behavior during the experiment. According to logical theories of meaning, the meaning of an expression can be formalized and encoded in truth conditions using a verification task. This task is a commonly used paradigm to test mental representations of meaning.

Hypotheses will be tested using the data-oriented computational model by Ramotowska [7], which is based on a three-parameter logistic function. I will analyze the different threshold values to determine if stereotypes have a predictable influence. The advantages of this model are clearly stated in their work; primarily, it allows for the separation of the threshold value, vagueness, and response error, and it accounts for variations in truth conditions between individuals. While response error is not a primary parameter for this analysis, it is fundamental to separate general errors from vagueness (errors due to the proximity to the threshold).

I will use a design similar to the one used in the study by Ramotowska et al., but with pictures instead of sentences. I have chosen a visual paradigm, even though it has been found that visual paradigms introduce more vagueness [8]. I believe that providing exact percentages in written sentences would make stereotyped individuals more aware of the inconsistencies in their responses across different contexts, thereby reducing the unconscious influence of stereotypes.

Participants will be presented with sentences such as "Many of the doctors are female" and "Many of the waiters are male". They will then see an image depicting people within that context, with a total number of individuals ranging from 10 to 30 per image, and a varying percentage of male and females. To ensure that male and female figures are easily recognizable, they will be color-coded (e.g., blue for men and pink for women).

A potential issue could be the "effect of amount" [12], however, as the range of amounts is relatively small in this experiment, this is unlikely to arise. Given the absence of an amount effect in this range, I decided to use between 10 and 30 objects per image to prevent participants from becoming consciously aware of the actual percentages they are evaluating (for the same reasons of above).

Since the context remains consistent (always involving

groups of men and women), if we find a difference between two clusters, that difference should be attributed to prior expectations. If these prior expectations can be predicted and align with the stereotypes a person holds, we can assume that the prior expectations are guided by those stereotypes.

Obviously, not having found any studies that tackle this specific problem (or a similar one), I tried to design the experiment using techniques and notions from the literature. Not being an expert in this field, I am aware that improvements could be made to make the experiment more rigorous, but I believe this serves as a good starting point.

6. References

[1] A. Schöller and M. Franke, "Semantic values as latent parameters: Surprising few & many," in *Semantics and linguistic theory*, 2015, pp. 143–162.

[2] T. Fernando and H. Kamp, "Expecting many," in *Semantics and linguistic theory*, 1996, pp. 53–68.

[3] H. H. Clark, "Words, the world, and their possibilities." 1991.

[4] S. E. Newstead and K. R. Coventry, "The role of context and functionality in the interpretation of quantifiers," *European Journal of Cognitive Psychology*, vol. 12, no. 2, pp. 243–259, 2000.

[5] B. Partee, "Many quantifiers," in *Proceedings of ESCOL*, vol. 5, 1988, pp. 383–402.

[6] L. Bott and I. A. Noveck, "Some utterances are underinformative: The onset and time course of scalar inferences," *Journal of memory and language*, vol. 51, no. 3, pp. 437–457, 2004.

[7] S. Ramotowska, J. Haaf, L. Van Maanen, and J. Szymanik, "Most quantifiers have many meanings," *Psychonomic bulletin & review*, vol. 31, no. 6, pp. 2692–2703, 2024.

[8] A. Sarafoglou, A. Giacobello, H. Godmann, T. Johnson, I. Visser, J. M. Haaf, and J. Szymanik, "A bayesian framework to study individual differences in semantic representations," 2024.

[9] S. Verheyen, S. Dewil, and P. Égré, "Subjectivity in gradable adjectives: The case of tall and heavy," *Mind & Language*, vol. 33, no. 5, pp. 460–479, 2018.

[10] S. Verheyen, E. Droeshout, and G. Storms, "Age-related degree and criteria differences in semantic categorization," *Journal of cognition*, vol. 2, no. 1, p. 17, 2019.

[11] A. Siyanova-Chanturia, F. Pesciarelli, and C. Cacciari, "The electrophysiological underpinnings of processing gender stereotypes in language," *PloS one*, vol. 7, no. 12, p. e48712, 2012.

[12] A. Schöller and M. Franke, "How many manys? exploring semantic theories with data-driven computational models," in *Proceedings of Sinn und Bedeutung*, vol. 20, 2016, pp. 622–639.